Interactive Fiction Game - Life With Adventure

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Objective and Requirements

The objective of this mini project is to design and create a text-based interactive fiction game. The project meets the following requirements:

- The player explores and interacts with a virtual world consisting of nine distinct places.
- Within these locations, the player performs actions such as collecting items, interacting with non-playable characters, and engaging in combat with enemies.
- I/O devices are also used, namely switches, the timer, button, and 7-Segment Display.
- Lastly, a non-trivial win condition is implemented.

The game allows the player to make choices and interact with the story through input and output of text, switches and a button. Each location presents choices that affect the player's progression and story outcome.

Solution

The project is mainly developed in C code and executed on the DTEK-V board. A RISC-V assembly implementation of `strcmp` was created and used within the C program to enable string comparison on the board. A terminal is used for the text-based input and output. In addition, the board's peripherals are utilized: switches 1-5 are used to perform actions such as attacking or viewing items in the inventory. The 7-segment display shows the player's health by displaying "HP 100" at the start and changing throughout the player's actions. KEY1 is used to perform various actions throughout the game, such as a jump in room 8.

A timer is used to keep track of the play time. A death system is implemented to end the game if the player health reaches 0. Conditions for every room are implemented, such as enemies in certain rooms which gradually increase in strength as the player progresses, to make the game fun and still challenging.

Rooms are represented using structs that contain descriptions, exits, enemies and non-playable characters. The player can move between rooms by choosing options via the switches. The player has attributes such as health, money, damage and an inventory. The inventory is accessible by toggling switch 0. Combat is implemented turn-based, where the player can attack, defend, heal or flee. Switches and button inputs are handled through both interrupts and polling.

Verification

The solution was verified by systematically testing the game on the DTEK-V board. Each requirement was individually tested to ensure it works as intended. For example, the terminal was used to verify correct text input and output, the switches and buttons were tested for triggering in-game actions. The room navigation was verified by testing all the room exits to confirm that the player moves to the intended locations. The timer was verified by measuring the time it took to play though the game. The display output was confirmed as well by observing that the 7-Segment Display correctly reflected the player's health. Additionally, edge cases were tested, such as attempting actions without required items and trying to pick up more items than possible.

Contributions

The workload was divided so that both of the parties were involved in programming as well as contributing to the design, debugging, and testing of the game. Frans primarily focused on the program logic and hardware-related implementation, such as interrupt handling, and managing switch inputs. He also contributed to the program flow and combat implementation.

Chandadeep focused more on the overall design, testing, and story implementation. She implemented the main game structure, including core gameplay functions for the rooms, and story actions. To ensure shared learning, responsibilities were rotated throughout the project.

Both members collaborated on debugging, testing on the board, and designing the game's flow, and room structure. Documentation and report writing were also done jointly.

Reflections

Working on this project was both challenging and rewarding. It allowed for the combination of hardware interaction with game design, which deepened the understanding of how software and hardware communicate. The importance of thorough planning, debugging, and team-working was also understood. Creating a text-based adventure game on the RISC-V board required creative thinking within technical limitations, and it was satisfying to see the final game run smoothly on real hardware.